Valid Starting City

AlgoExpert · Medium · Greedy Algorithms

The problem

- Circular route; city i has fuel[i] gallons and is distances[i] miles from the next.
- Return the unique start index allowing a full loop.
- The start is the point after the lowest fuel balance.

Intuition

- The valid start is right after the point where the running fuel balance is lowest.

Brute force vs optimal

- Brute force: simulate from every city → $O(n^2)$
- Track running fuel; start where it's lowest → $O(n)$
- Guaranteed exactly one valid city

Approach

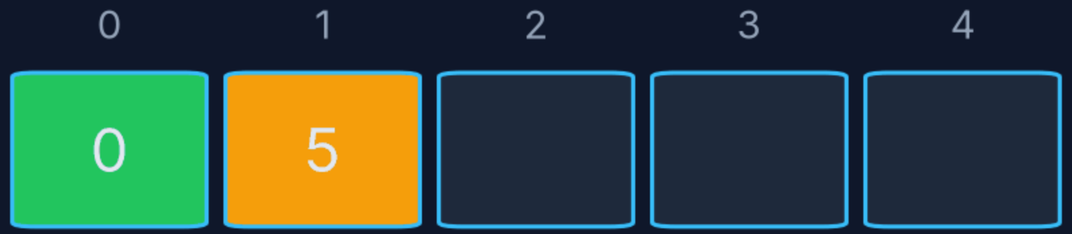
- 1. Track miles remaining around the loop
- 2. Find the city with the minimum balance
- 3. That city is the start

Worked example

distance to next



miles left on arrival (starting at city 0)



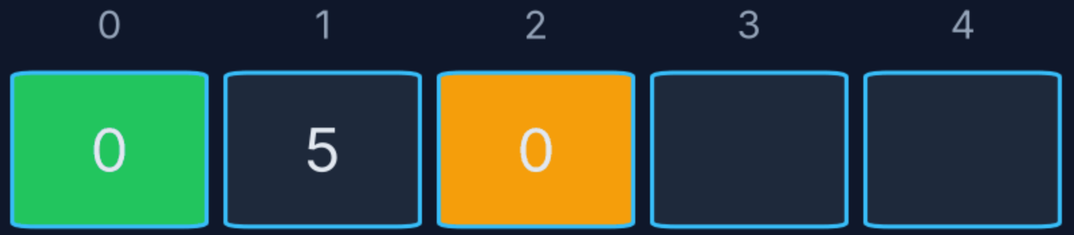
Arrive at city 1 with 5 miles; lowest so far at city 0

Worked example

distance to next



miles left on arrival (starting at city 0)



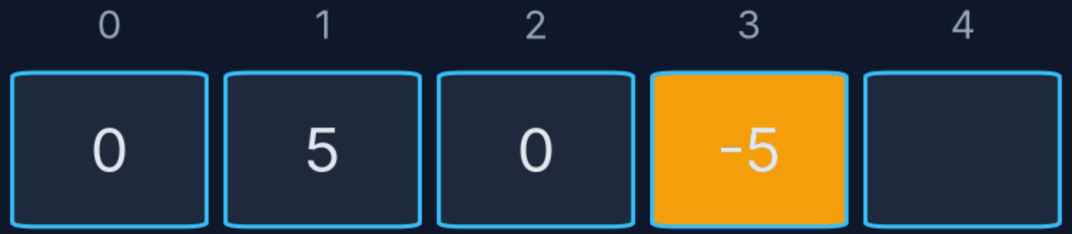
Arrive at city 2 with 0 miles; lowest so far at city 0

Worked example

distance to next



miles left on arrival (starting at city 0)



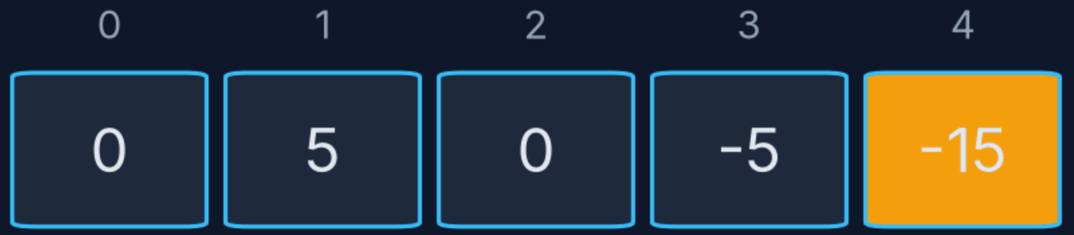
Arrive at city 3 with -5 miles; lowest so far at city 3

Worked example

distance to next



miles left on arrival (starting at city 0)



Arrive at city 4 with -15 miles; lowest so far at city 4

Java solution

```java
public class ValidStartingCity {
    public static int validStartingCity(int[] distances, int[] fuel, int mpg) {
        int miles = 0, minMiles = 0, start = 0;
        for (int i = 1; i < distances.length; i++) {
            miles += fuel[i - 1] * mpg - distances[i - 1];
            if (miles < minMiles) { minMiles = miles; start = i; }
        }
        return start;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Fuel is gallons × mpg
- Start at the minimum *arrival* point
- Input guarantees a solution

Interview tips & follow-ups

- Same as LeetCode Gas Station
- Explain why the minimum point works

Recap

- The valid start is right after the point where the running fuel balance is lowest.
- Complexity: O(n)